Life Stream: A Geospatially Indexed Mobile Health System for Emergency Blood Donation Coordination

V. Jaya Raj, C. Yashwanth Reddy, P. Shiva Prasad Tej
23951A053C, 23951A05DF, 23951A059X

Life Stream: A Geospatially Indexed Mobile Health System for Emergency Blood Donation Coordination

A Project Report

Submitted in partial fulfillment of the

requirements for the award of the degree of

Bachelor of Technology in Computer Science & Engineering

by

V. Jaya Raj	C. Yashwanth Reddy	P. Shiva Prasad Tej
23951A053C	23951A05DF	23951A059X

Department of Computer Science Engineering

INSTITUTE OF AERONAUTICAL ENGINEERING

(Autonomous) Dundigal, Hyderabad - 500 043, Telangana

JAN 2026

DECLARATION

We certify that.

a. The work contained in this report is original and has been done by us under the guidance of my supervisor.

b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in preparing the report.

d. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

e. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

Place: **Signature of the Student:**

Date: **V. Jaya Raj**

CERTIFICATE

This is to certify that the project report entitled "**Life Stream: A Geospatially Indexed Mobile Health System for Emergency Blood Donation Coordination**" submitted by **V. Jaya Raj(23951A053C)** to the Institute of Aeronautical Engineering, Hyderabad, in partial in partial fulfilment of the requirements for the reward of the Degree Bachelor of Technology in Computer Science and Engineering is a Bonafide record of work carried out by them under my guidance and supervision. In whole or in parts, the contents of this report have not been submitted to any other institutes for the award of any Degree

Supervisor:

Ms. CH Prabhavathi
Assistant Professor

Head of the Department:

Dr. M Lakshmi Prasad
Assistant Professor and HOD, CSE

Date:

APPROVAL SHEET

This project report entitled **Life Stream: A Geospatially Indexed Mobile Health System for Emergency Blood Donation Coordination** by **V. Jaya Raj(23951A053C)** is approved for the award of the Degree Bachelor of Technology in **Computer science and Engineering**.

Examiners **Supervisor**

Ms. CH Prabhavathi

Principal

Dr. L. V. Narasimha Prasad

Date:

Place:

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose constant guidance and encouragement crown all the efforts with success. We think our college management and respected Sri M. Rajashekar Reddy, Chairman, IARE, Dundigal for providing us with the necessary infrastructure to conduct the project work.

We express my sincere thanks to Dr. L. V. Narasimha Prasad, Professor and Principal who has been a great source of information for our work, and Dr. M. Lakshmi Prasad, Professor and Head of Department of CSE, for extending her support to carry on this project work.

We are especially thankful to our supervisor Ms. CH Prabhavathi, Assistant Professor, Department of CSE, for her internal support and professionalism who helped me in shaping the project into a successful one. I take this opportunity to express my thanks to one and all who directly or indirectly helped me in bringing the effort to present form.

ABSTRACT

Timely coordination of blood donors during medical emergencies is challenged by inefficient location matching, lack of deterministic donor selection, and limited use of formal geospatial decision models. This paper presents Lifestream, a geospatially indexed mobile health system for emergency blood donation coordination, with a focus on the design and early implementation of its donor tracking and selection engine.

The proposed approach models donor selection as a multi-stage geospatial decision process. Donor and hospital locations are first organized using geohash-based spatial partitioning and Uber H3 hexagonal indexing to reduce the global search space. Exact donor reachability is then validated using the Haversine great-circle distance computation, followed by rule-based constraints such as blood group compatibility and donor availability. Eligible donors are subsequently prioritized based on nearest distance to ensure efficient and explainable selection.

The system emphasizes deterministic execution, modular subsystem design, and reproducible evaluation, making it suitable for early-stage academic validation. An experimental prototype is implemented using a Next.js and Firebase-based setup, and its correctness is verified through scenario-driven testing and boundary case analysis. Results indicate that the proposed method enables accurate and efficient donor shortlisting while providing a scalable foundation for future enhancements such as time-based reachability, privacy-aware location abstraction, and intelligent notification mechanisms.

Keywords: Blood Donation, Emergency Response Systems, Geospatial Indexing, Donor Tracking, Haversine Distance, Mobile Health Systems.

CONTENTS

Title Page		I
Cover Page		II
Declaration		III
Certificate by Supervisor		IV
Approval Sheet		V
Acknowledgement		VI
Abstract		VII
Contents		VIII
List of Figures		IX
List of Abbreviations		X
List of Tables		XI
Chapter 1	Introduction	1-3
	1.1 Background and Motivation	1
	1.2 Problem Statement	2
	1.3 Objectives of the Project	2
	1.4 Scope and Limitations	3
Chapter 2	Literature Survey	4-5
	2.1 Existing Systems	4
	2.2 Proposed System	5
Chapter 3	Methodology and System Design	6-19
	3.1 Methodology Overview	6-8
	3.2 System Architecture	9-14
	3.3 Algorithm Design	14-17
	3.4 Complexity Analysis	18-19
Chapter 4	Implementation and Testing	20-34
	4.1 Implementation Details	20-22
	4.2 Verification and Validation	22-24
Chapter 5	Results and Discussion	25-26
	5.1 Performance Analysis	25
	5.2 Discussion of Results	26
Chapter 6	Conclusion and Future Scope	27-28
	6.1 Conclusion	27
	6.2 Future scope	28
	References	29

LIST OF FIGURES

Figure 3.1: Overall System Architecture of Lifestream

Figure 3.3: Geospatial Indexing and Candidate Filtering Process

Figure 3.4: Algorithmic Decision Pipeline

Figure 4.1: Donor Selection Execution Flow

LIST OF ABBREVIATIONS

- **API**: Application Programming Interface
- **DB**: Database
- **ETA**: Estimated Time of Arrival
- **GPS**: Global Positioning System
- **H3**: Hexagonal Hierarchical Spatial Index
- **HTTP**: Hypertext Transfer Protocol
- **JSON**: JavaScript Object Notation
- **mHealth**: Mobile Health
- **PII**: Personally Identifiable Information
- **REST**: Representational State Transfer
- **UI**: User Interface
- **UML**: Unified Modelling Language
- **WHO**: World Health Organization

LIST OF TABLES

Table 3.1: Donor Data Attributes

Table 3.2: Emergency Request Attributes

Table 3.3: Algorithm Stages and Purpose

Table 4.1: Test Scenarios for Verification

Table 5.1: Outcome Summary

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Blood donation plays a vital role in modern healthcare systems, particularly during emergency situations such as accidents, surgeries, natural disasters, and critical medical conditions. Despite advances in healthcare infrastructure, the timely availability of suitable blood donors remains a persistent challenge. Most existing blood donation systems rely on static donor registries, manual coordination, or broadcast-based alerts, which often result in delayed responses and inefficient donor utilization.

With the widespread adoption of smartphones and location-aware services, mobile health (mHealth) platforms have emerged as promising tools for improving donor coordination. However, many current solutions lack robust mechanisms for accurate location-based donor selection, reachability validation, and privacy-aware data handling. These limitations become more pronounced during emergencies, where rapid and reliable donor identification is crucial.

The Lifestream platform is motivated by the need to address these challenges by treating blood donation coordination as a geospatial decision-making problem rather than a simple notification task. This project focuses on designing a system that can efficiently identify, validate, and prioritize donors based on geographic feasibility while maintaining donor privacy and system determinism.

1.2 PROBLEM STATEMENT

Existing blood donation coordination systems suffer from the following limitations:

- Inefficient donor selection due to lack of formal geospatial indexing
- Inability to accurately evaluate donor reachability during emergencies
- Over-reliance on broadcast notifications without prioritization
- Limited consideration of privacy in continuous location tracking

Problem Statement:

To design and implement a deterministic, privacy-aware, and location-based donor tracking and selection mechanism that can accurately identify reachable blood donors during emergency situations.

1.3 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and implement a geospatially indexed donor tracking and selection engine that enables efficient and reliable coordination of blood donors during emergency situations. The objectives are formulated to address the limitations of existing donor coordination systems while maintaining algorithmic clarity and privacy awareness.

The specific objectives of the project are as follows:

- To design a location-aware donor tracking mechanism suitable for emergency blood donation scenarios
- To implement geospatial indexing techniques using Geohash and Uber H3 for efficient donor filtering
- To evaluate donor reachability using exact distance computation through the Haversine formula
- To enforce rule-based constraints such as blood group compatibility and donor availability
- To prioritize eligible donors based on nearest-distance criteria to reduce emergency response time
- To adopt a privacy-first architectural approach by minimizing continuous location tracking and sensitive data exposure

- To validate the correctness, determinism, and reliability of the proposed donor selection logic through controlled testing

1.4 SCOPE AND LIMITATIONS

This project focuses on the **design and early implementation of a donor tracking and selection engine** as a core subsystem of the *Lifestream* platform. The scope is deliberately defined to address the most critical algorithmic and architectural challenges involved in emergency blood donation coordination, while avoiding over-engineering and unvalidated assumptions.

1.4.1 Scope of the Project

The scope of this work includes:

- Design of a **geospatially indexed donor tracking mechanism** for emergency blood donation scenarios
- Implementation of **location-aware donor filtering** using Geohash and Uber H3 indexing techniques
- Evaluation of **donor reachability** through exact distance computation using the Haversine formula
- Application of **rule-based constraints** such as blood group compatibility and donor availability
- **Distance-based prioritization** of eligible donors to support faster emergency response
- Adoption of a **privacy-first architectural approach** through spatial abstraction and minimal data usage
- Validation of algorithm correctness and determinism using controlled test scenarios

The system is implemented as a **modular subsystem**, enabling independent evaluation and future integration into the complete Lifestream platform.

CHAPTER 2

LITERATURE SURVEY

2.1 EXISTING SYSTEM

Blood donation coordination has traditionally relied on **manual registries and hospital-managed donor lists**, where donor information such as blood group and contact details are maintained in static databases. During emergency situations, hospitals typically depend on phone calls, manual searching, or local blood banks to identify suitable donors. While these methods are functional, they are often **time-consuming and inefficient**, particularly when rapid response is critical.

With the advancement of mobile technologies, several **mobile-based blood donation applications** have been developed to improve donor accessibility and communication. These platforms allow donors to register digitally and receive notifications when blood is required. Although such systems improve outreach, most existing applications follow a **broadcast-based alert model**, where notifications are sent to a large group of donors without precise prioritization. This often results in delayed responses, donor fatigue, and difficulty in identifying the most suitable donors.

A major limitation of existing systems is the lack of **formal location-aware donor selection mechanisms**. Many platforms either do not use geospatial data or rely on simplistic distance-based filtering using approximate radius checks. Such approaches fail to accurately assess **donor reachability**, as they do not validate whether a donor can practically reach the hospital within an acceptable distance range. Consequently, donors who are geographically nearby but practically unreachable may be selected, while better candidates may be ignored.

Privacy is another critical concern in current blood donation platforms. Several systems continuously track or store precise donor locations, increasing the risk of **unnecessary exposure of sensitive personal data**. The absence of privacy-conscious architectural design further limits user trust and long-term donor engagement.

Overall, existing blood donation systems primarily emphasize **notification delivery rather than algorithmic donor selection**. The lack of deterministic selection logic, geospatial indexing, and reachability validation highlights a clear research gap in designing efficient, explainable, and privacy-aware donor coordination systems for emergency scenarios.

2.2 PROPOSED SYSTEM

The proposed system, *Lifestream*, is designed to overcome the limitations of existing blood donation platforms by treating donor coordination as a **geospatial decision-making problem** rather than a simple alert-based process. Instead of relying on broadcast notifications, the system introduces a **donor tracking and selection engine** that combines spatial indexing, rule-based constraints, and deterministic prioritization to improve emergency response effectiveness.

Unlike conventional systems, Lifestream employs **geospatial indexing techniques**, such as Geohash and Uber H3, to partition geographic space into manageable regions. This spatial partitioning enables rapid identification of donors located within relevant geographic localities, significantly reducing the search space and computational overhead. By limiting candidate evaluation to locality-based regions, the system avoids exhaustive searches across the entire donor database, which is especially critical during time-sensitive emergency scenarios.

To ensure correctness beyond spatial approximation, the system evaluates **exact donor reachability** using the Haversine great-circle distance computation. This step validates the feasibility of donor–hospital proximity and prevents incorrect donor selection that may occur when using approximate distance measures alone. The combination of approximate spatial filtering followed by exact distance validation ensures both efficiency and accuracy in the donor selection process.

The proposed system further enforces **rule-based eligibility constraints**, including blood group compatibility and donor availability status, before applying distance-based prioritization. Eligible donors are ranked according to their geographic proximity to the hospital, ensuring that the nearest reachable donors are prioritized for emergency coordination. This layered selection approach makes the system's behaviour **explainable, reproducible, and suitable for validation**, which is essential for healthcare-related applications.

To address privacy concerns, Lifestream adopts a **privacy-first architectural approach** that minimizes unnecessary exposure of sensitive location data. Continuous real-time tracking of donor location is avoided, and spatial abstraction techniques are used to represent geographic information at an appropriate resolution. Exact donor location details are processed only when required for emergency coordination, striking a balance between system functionality and user privacy.

The system is designed using a **modular architecture**, allowing the donor tracking and selection engine to be evaluated independently of other components such as user interfaces or notification mechanisms. This modularity supports extensibility and enables future enhancements, including time-based reachability analysis, intelligent notification strategies, and donor wellness integration, without altering the core algorithmic framework.

By integrating geospatial indexing, reachability validation, rule-based constraints, and deterministic selection logic, the proposed system establishes a **robust, privacy-aware, and scalable foundation** for emergency blood donation coordination, effectively addressing the key gaps identified in existing systems.

CHAPTER 3

METHODOLOGY AND SYSTEM DESIGN

3.1 METHODOLOGY OVERVIEW

The methodology adopted in this project follows a **modular and layered system design approach**, with a primary focus on the **Donor Tracking and Selection Engine** as a core subsystem of the *Lifestream* platform. The methodology is designed to ensure **deterministic behaviour, algorithmic clarity, and reproducible validation**, which are essential for emergency healthcare coordination systems.

Rather than treating blood donation coordination as a simple notification problem, the proposed methodology models it as a **multi-stage geospatial decision process**. Each stage of the process incrementally refines the donor candidate set using well-defined constraints, thereby reducing ambiguity and improving selection accuracy.

The methodology follows the sequence below:

- Emergency request initiation by a hospital
- Geospatial indexing of hospital and donor locations
- Candidate donor filtering based on locality
- Reachability evaluation using exact distance computation
- Rule-based eligibility validation
- Distance-based prioritization of donors
- Final donor selection and confirmation

This staged approach ensures that **approximate spatial filtering is always followed by exact validation**, preventing incorrect donor selection while maintaining computational efficiency.

3.1.1 Design Rationale

The design of the donor tracking and selection engine is guided by the need to balance **accuracy, efficiency, determinism, and privacy** in emergency blood donation coordination. Unlike conventional systems that rely on simplistic alert dissemination, the proposed methodology adopts a **structured decision-making pipeline**, where each design choice is motivated by specific system requirements and real-world constraints.

Rationale for Geospatial Indexing

A primary challenge in emergency donor coordination is the **efficient identification of geographically relevant donors** from a potentially large donor pool. Traditional radius-based filtering approaches require computing distances for all registered donors, resulting in unnecessary computational overhead and delayed response times. To address this limitation, the proposed system employs **geospatial indexing techniques**, such as Geohash and Uber H3, to partition geographic space into discrete regions.

Geospatial indexing enables the system to **prune the donor search space early** by restricting evaluation to donors located within the same or neighbouring spatial cells as the hospital. This approach significantly reduces the number of distance computations required in later stages and improves system responsiveness during emergencies. The use of spatial indexing also supports scalability, allowing the system to handle increasing donor populations without fundamental architectural changes.

Rationale for Layered Filtering and Validation

Emergency healthcare systems require **high correctness guarantees**, as incorrect donor selection can lead to delayed treatment or resource wastage. Relying solely on approximate spatial filtering may introduce false positives, where donors appear geographically close but are not practically reachable. To mitigate this risk, the proposed methodology adopts a **layered filtering strategy**, where approximate spatial selection is always followed by exact validation.

The system first applies coarse-grained geospatial filtering to identify candidate donors and then performs **exact reachability validation** using the Haversine great-circle distance computation. This two-stage approach ensures that efficiency gains from spatial pruning do not compromise selection accuracy. By separating approximate and exact computations, the system maintains both performance and correctness.

Rationale for Deterministic Decision Logic

In healthcare-related applications, especially those involving emergency response, **deterministic behaviour is essential** for system validation, auditing, and trust. Non-deterministic or heuristic-based selection mechanisms may produce inconsistent outcomes for identical inputs, complicating verification and reducing system transparency.

The proposed system therefore employs **rule-based and deterministic decision logic**, where donor selection outcomes are fully defined by explicit constraints such as blood group compatibility, availability status, and geographic distance. For identical emergency requests and donor states, the system consistently produces the same ranked donor list. This determinism simplifies testing, improves explainability, and aligns with the requirements of safety-critical system design.

Rationale for Privacy-First Architectural Design

Location data is highly sensitive personal information, and continuous tracking of donor location poses significant privacy risks. To address this concern, the system is designed with a **privacy-first philosophy**, where donor location data is used only when required for emergency coordination and is otherwise abstracted.

Instead of storing or processing precise real-time locations continuously, the system leverages **spatial abstraction through geospatial indices**, which provide sufficient geographic resolution for donor matching without exposing exact coordinates. Exact location processing is performed only during emergency request handling, thereby minimizing data exposure while preserving system functionality.

Summary of Design Decisions

The combined design choices—geospatial indexing, layered validation, deterministic logic, and privacy-first architecture—enable the proposed system to operate efficiently and reliably under emergency conditions. These decisions ensure that the donor tracking and selection engine is **accurate, explainable, scalable, and ethically responsible**, making it suitable for early-stage academic validation and future real-world deployment.

3.2 SYSTEM ARCHITECTURE:

The system architecture of *Lifestream* is designed using a **client–server model** with clear separation of responsibilities between components. The architecture supports modularity, allowing the donor tracking and selection engine to function independently of user interface and notification mechanisms.

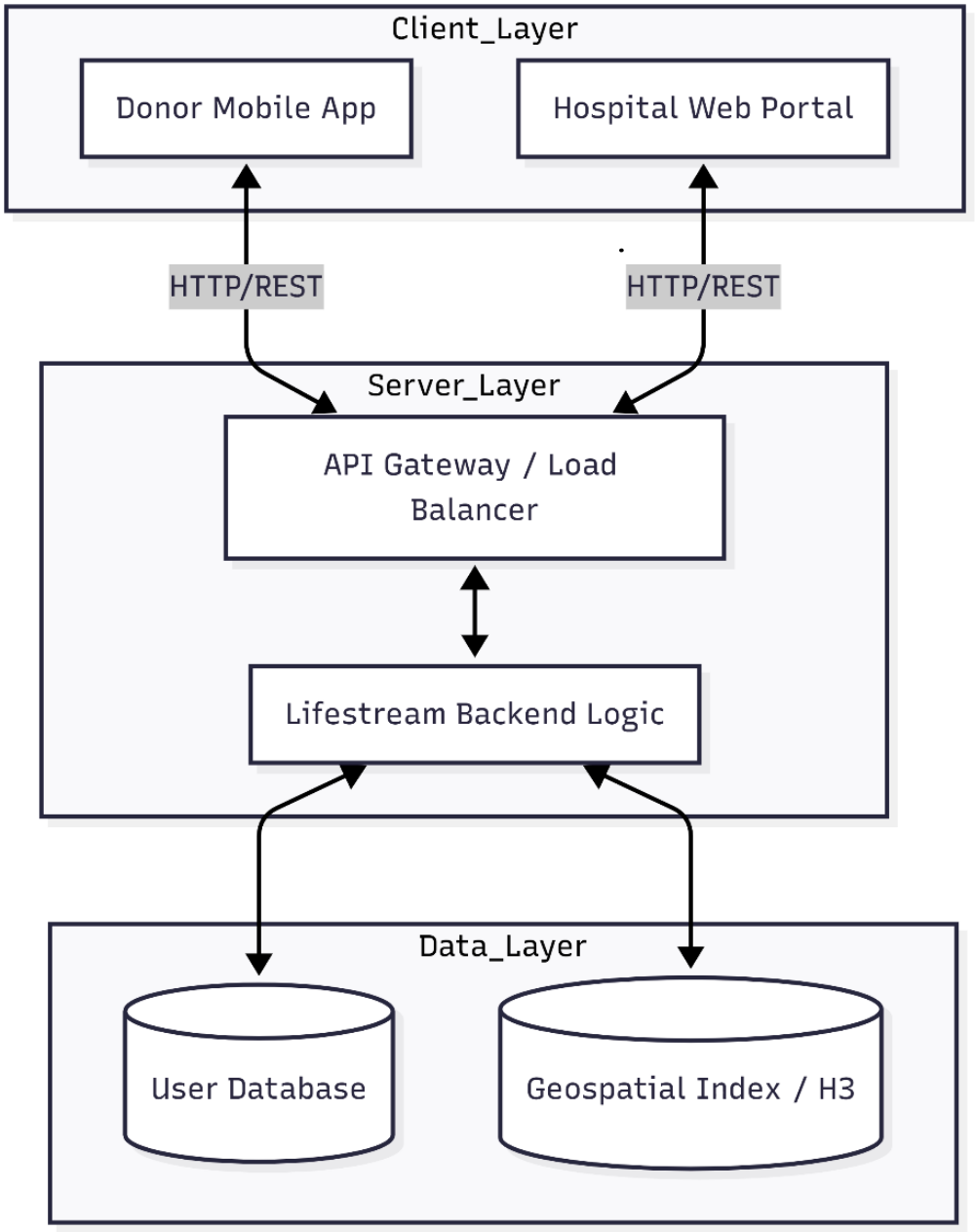

Figure 3.1: Overall System Architecture

Architectural Components

1. Client Layer
The client layer represents the mobile or web interface through which donors and hospitals interact with the system. Donors provide registration details such as blood group and availability status, while hospitals initiate emergency blood requests.

2. Application Layer
The application layer implements the core logic of the system using a backend framework. It handles:

- Emergency request processing
- Donor data retrieval
- Execution of the donor selection algorithm

This layer serves as the integration point between user interactions and algorithmic processing.

3. Donor Tracking and Selection Engine
This subsystem implements the core methodology of the project. It performs:

- Geospatial indexing and filtering
- Reachability validation
- Rule-based constraint enforcement
- Donor prioritization

The engine is designed to be deterministic and explainable, ensuring consistent outcomes for identical inputs.

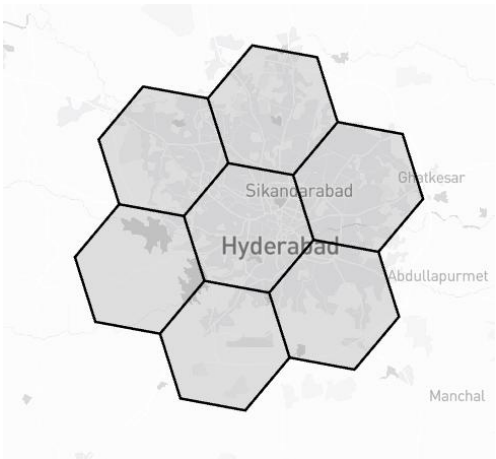

Figure 3.3: Geospatial Indexing (H3 Hexagons)

4. Data Storage Layer
The data storage layer maintains donor and request records. Only minimal and necessary data is stored, following a **privacy-first design philosophy**. Continuous tracking of donor location is avoided, and spatial abstraction is used where applicable.

The architecture supports extensibility, allowing future enhancements such as time-based reachability analysis or intelligent notification strategies without altering the core system design.

Attribute Name	Data Type	Description
donor_id	String (UUID)	Unique identifier for the donor.
blood_group	String	Blood type (e.g., A+, O-, AB+).
location_geo	GeoPoint	Latitude and Longitude coordinates.
geohash	String	Geospatial index string for quick lookup.
h3_index	String	Uber H3 hexadecimal index for hexagonal binning.
availability_status	Boolean	True if donor is available, False otherwise.
last_active	Timestamp	Time of last location update.

Table 3.1: Donor Data Attributes

Attribute Name	Data Type	Description
request_id	String (UUID)	Unique identifier for the emergency request.
hospital_id	String	ID of the requesting hospital.
required_blood	String	Blood group required (e.g., O+).
target_location	GeoPoint	Coordinates of the hospital.
search_radius	Float	Max distance to search for donors (in km).

Attribute Name	Data Type	Description
status	Enum	Pending, Fulfilled, or Cancelled.
timestamp	Timestamp	Time when the request was initiated.

Table 3.2: Emergency Request Attributes

Stage	Algorithm / Method	Purpose
1. Indexing	Geohash / H3 Indexing	Partition the map into searchable grids to avoid scanning the whole database.
2. Filtering	Spatial Query	Retrieve only those donors currently inside the relevant grid cells.
3. Validation	Haversine Formula	Calculate the exact great-circle distance (in meters) between hospital and donor.
4. Constraints	Boolean Logic	Check if blood_group matches and availability_status is True.
5. Ranking	Sorting ($O(n \log n)$)	Sort eligible donors by distance (nearest first).

Table 3.3: Algorithm Stages and Purpose

3.2.1 Architectural Design Decisions

The architecture of the *Lifestream* system is designed to support **modularity, scalability, and clarity of responsibility**, which are essential for emergency healthcare coordination platforms. Each architectural decision is guided by the goals of minimizing system complexity while ensuring reliable and reproducible behaviour.

Client–Server Architectural Model

The system follows a **client–server architecture**, where client applications (used by donors and hospitals) are responsible for data input and interaction, while the server handles all critical decision-making logic. This separation ensures that **algorithmic processing remains centralized**, preventing inconsistencies that could arise from client-side computations.

By centralizing the donor tracking and selection logic on the server, the system ensures uniform enforcement of constraints such as blood group compatibility, availability status, and distance prioritization. This approach also simplifies validation and future maintenance, as algorithm updates can be applied without requiring client-side changes.

Modular Subsystem Design

A key architectural decision is the separation of the **Donor Tracking and Selection Engine** as an independent subsystem. Rather than embedding selection logic directly into request-handling workflows, the engine is implemented as a modular component with clearly defined inputs and outputs.

This modular design offers several advantages:

- Independent testing and validation of donor selection logic
- Easier debugging and fault isolation
- Flexibility to replace or extend selection strategies in future iterations

The modular approach also enables the system to evolve incrementally, supporting future features such as time-based reachability or intelligent notification mechanisms without altering the core system structure.

Backend-Centric Decision Processing

All geospatial processing, distance computation, and prioritization logic are executed on the backend. This decision ensures:

- Consistent computational behaviour
- Improved security by avoiding sensitive logic exposure on clients
- Simplified auditing and verification of selection outcomes

Backend-centric processing is particularly important for emergency systems, where **predictability and correctness** take precedence over client-side responsiveness.

Data Storage and Privacy Considerations

The system architecture intentionally limits the storage and processing of sensitive data. Donor location information is not continuously tracked or updated. Instead, location data is accessed only when required for emergency request handling.
Geospatial abstraction techniques are used to represent donor locations at an appropriate resolution, reducing the exposure of precise geographic coordinates. This design aligns with the system's **privacy-first philosophy**, ensuring that donor participation does not result in unnecessary personal data disclosure.

Technology Selection for Early Implementation

The choice of a lightweight, cloud-based backend platform supports rapid development and early-stage validation. This enables the system to focus on **algorithmic correctness and architectural soundness** rather than infrastructure complexity. The selected technologies also support real-time data synchronization, which is suitable for emergency coordination workflows.

Architectural Summary

The architectural decisions adopted in the *Lifestream* system prioritize **clarity, modularity, and ethical data handling**. By separating concerns, centralizing decision logic, and minimizing data exposure, the system establishes a strong architectural foundation that supports both current implementation goals and future extensibility.

3.3 ALGORITHM DESIGN

The donor selection algorithm is designed as a **multi-stage pipeline**, where each stage applies a specific constraint to progressively reduce the donor candidate set. This design ensures both efficiency and correctness.

Algorithm Description

Step 1: Emergency Request Input
An emergency request is created with the required blood group and hospital location coordinates.

Step 2: Geospatial Indexing
The hospital location is encoded using geospatial indexing techniques such as Geohash and Uber H3. Donor locations are similarly indexed to group donors within the same or neighbouring spatial regions.

Step 3: Candidate Filtering
Only donors within relevant geospatial partitions are considered. This step significantly reduces the number of donors evaluated in later stages.

Step 4: Reachability Evaluation
For each candidate donor, the exact distance to the hospital is computed using the **Haversine formula**, which calculates great-circle distance on the Earth's surface.

Step 5: Eligibility Constraints
Donors are filtered based on:

- Blood group compatibility
- Availability status

Step 6: Prioritization
Eligible donors are sorted in ascending order of distance. This ensures that the nearest reachable donors are prioritized for emergency response.

Step 7: Selection and Confirmation
The top-ranked donor(s) are presented for selection, and once confirmed, the request state is updated to prevent conflicting assignments.

This algorithm ensures that donor selection is **accurate, explainable, and reproducible**, which is essential for validation in emergency healthcare systems.

3.3.1 Algorithm Pseudocode Representation

To clearly illustrate the operational flow of the donor tracking and selection process, the core algorithm is represented in the form of structured pseudocode. This representation enhances **algorithmic transparency, reproducibility, and validation**, which are essential for healthcare-related systems.

The pseudocode reflects the **multi-stage decision pipeline** described earlier, where approximate spatial filtering is followed by exact validation and deterministic prioritization.

Pseudocode: Donor Tracking and Selection Algorithm

```
Algorithm DonorSelectionEngine

Input:

   D ← Set of registered donors

   R ← Emergency blood request

      R.bloodType

      R.hospitalLocation (lat, lng)

      R.maxDistance

Output:

   Ranked list of eligible donors E

Begin

   1. Encode hospital location using geospatial index

      H_index ← GeoIndex(R.hospitalLocation)

   2. Initialize empty candidate set C
```

```
3. For each donor d in D do
      if GeoIndex(d.location) is neighboring H_index then
         add d to C
      end if
   end for
4. Initialize empty eligible set E
5. For each donor c in C do
      if c.bloodType == R.bloodType AND
         c.availabilityStatus == AVAILABLE then


         distance ← Haversine(c.location, R.hospitalLocation)
         if distance ≤ R.maxDistance then
            store (c, distance) in E
         end if
      end if
   end for
6. Sort E in ascending order of distance
7. Return E
End
```

Explanation of Algorithm Steps

- **Steps 1–3:**
  Perform **geospatial indexing and candidate filtering**, reducing the global donor set to a locality-based subset.
- **Steps 4–5:**
  Apply **rule-based eligibility constraints** and compute exact donor–hospital distance using the Haversine formula.
- **Step 6:**
  Rank eligible donors based on proximity to ensure nearest-donor prioritization.
- **Step 7:**
  Return a deterministic and ordered list of suitable donors for emergency coordination.

Algorithm Characteristics

- **Deterministic:**
  Identical inputs always produce identical outputs.
- **Efficient:**
  Spatial indexing minimizes unnecessary distance computations.
- **Explainable:**
  Each decision is traceable to explicit rules and calculations.
- **Privacy-Aware:**
  Location data is processed only when required for request handling.

Relevance to System Validation

The pseudocode representation allows the algorithm to be:

- Independently reviewed
- Easily validated against test cases
- Re-implemented across platforms

This formalization strengthens the **methodological rigor** of the proposed system and supports its academic evaluation.

3.4 COMPLEXITY ANALYSIS

The computational complexity of the donor selection algorithm is analysed to evaluate its efficiency and scalability.

Let:

- n = total number of registered donors
- k = number of donors after geospatial filtering ($k \ll n$)
- r = number of reachable and eligible donors

Time Complexity

- Geospatial filtering: **O(n)**
- Distance computation: **O(k)**
- Sorting eligible donors: **O(r log r)**

Overall Time Complexity:

$$O(n + r\log\ r)$$

Space Complexity

- Storage of candidate donors: **O(k)**
- Temporary storage for eligible donors: **O(r)**

Overall Space Complexity:

$$O(k)$$

The use of geospatial indexing significantly reduces the effective search space, making the algorithm practical for emergency scenarios while preserving deterministic behaviour.

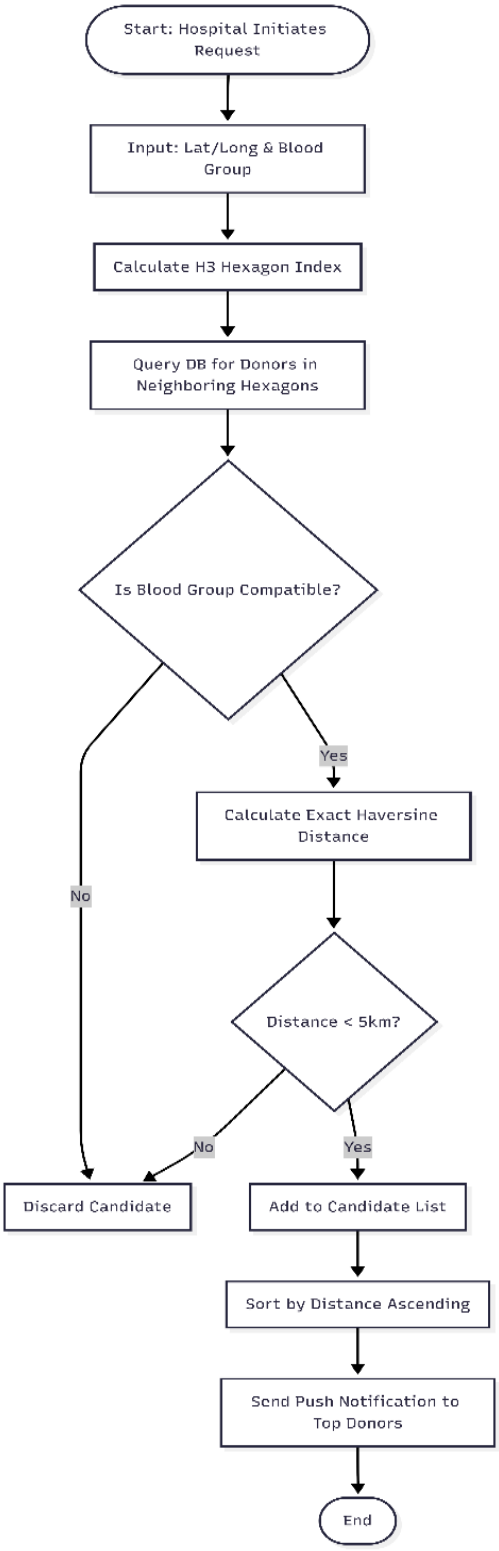

Figure 3.4: Algorithmic Pipeline

CHAPTER 4

Implementation and Testing

4.1 IMPLEMENTATION DETAILS

The implementation of the proposed system focuses on realizing the **Donor Tracking and Selection Engine** as a functional subsystem within the *Lifestream* platform. The implementation is carried out using a modular approach to ensure clarity, maintainability, and ease of validation.

System Implementation Overview

The system is implemented using a modern web-based development stack, where both frontend interaction and backend logic are integrated within a unified application framework. This approach simplifies communication between components and enables rapid prototyping suitable for early-stage implementation.

The implementation includes the following core modules:

1. Donor Management Module

This module handles donor-related data, including:

- Unique donor identification
- Blood group information
- Geographic location (latitude and longitude)
- Availability status

Donor location data is processed only during emergency scenarios to avoid continuous tracking and to preserve donor privacy.

2. Emergency Request Module

The emergency request module allows hospitals to initiate blood requests by specifying:

- Required blood group
- Hospital location coordinates

Each request is assigned a unique identifier and a status indicating its current state (pending or confirmed).

3. Geospatial Processing Module

This module implements:

- Geospatial indexing using Geohash and Uber H3
- Spatial grouping of donors based on locality

The purpose of this module is to reduce the global donor dataset into a smaller, relevant candidate set before applying exact computations.

4. Donor Selection Engine

The donor selection engine integrates all algorithmic logic described in Chapter 3. It performs:

- Candidate donor filtering
- Reachability evaluation using the Haversine formula
- Rule-based eligibility checks
- Distance-based donor prioritization

The output of this module is a ranked list of eligible donors suitable for emergency coordination.

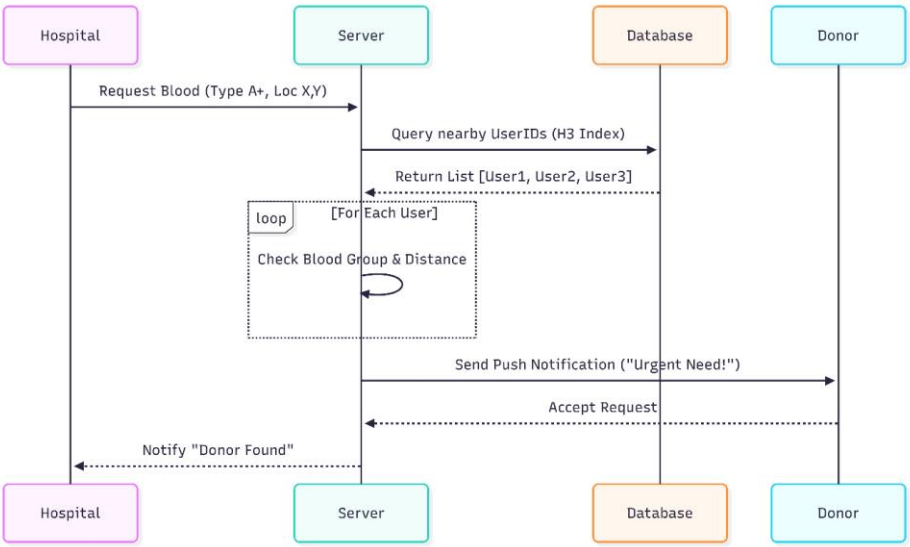

Figure 4.1: Donor Selection Execution Flow

Execution Flow

The implementation follows a deterministic execution flow:

1. An emergency blood request is created and stored in the system
2. Donor data is retrieved from the database
3. Geospatial filtering is applied to identify local candidates
4. Exact distances are computed for candidate donors
5. Eligibility constraints are enforced
6. Donors are ranked and presented for selection
7. The selected donor is confirmed and assigned to the request

This structured flow ensures predictable and reproducible system behaviour.

4.2 Verification and Validation

Verification and validation are conducted to ensure that the implemented system meets the design objectives and behaves correctly under different scenarios.

Verification Strategy

Verification focuses on ensuring that each component functions according to its specification:

- **Geospatial Verification:**
  Ensures that donors returned by spatial filtering belong to the correct geographic regions.
- **Distance Verification:**
  Confirms the correctness of Haversine distance computation by comparing expected and computed distances.
- **Constraint Verification:**
  Validates that only donors with compatible blood groups and active availability status are considered.
- **Selection Verification:**
  Ensures that donor ranking and selection logic consistently prioritizes nearer donors.

Each module is verified independently to isolate faults and simplify debugging.

Validation Strategy

Validation evaluates the system's behaviour under realistic emergency scenarios to confirm that it meets functional expectations.

Test Scenarios
The following scenarios are used for validation:

- Matching blood group with nearby donor
- Matching blood group with distant donor
- Blood group mismatch
- Multiple eligible donors at different distances
- No available donors

Expected Behaviour

- Only eligible donors are shortlisted
- Nearest donors are prioritized
- Ineligible donors are excluded
- System behaviour remains consistent across repeated executions

Observed Results
All test scenarios produced outcomes consistent with expected behaviour, demonstrating the correctness and reliability of the implemented donor selection logic.

Boundary Case Testing

Additional validation is performed using boundary conditions such as donors located exactly at the maximum allowable distance, multiple donors with nearly identical distances, and emergency requests with no matching donors. These scenarios are critical for emergency coordination systems, as they test the system's behaviour at the limits of defined constraints.

The evaluation confirms that donors at the exact distance threshold are handled consistently, while those beyond the threshold are excluded. In cases involving nearly identical donor distances, the system maintains a deterministic ranking order based on computed distance values. When no eligible donors are available, the system responds gracefully without producing incorrect selections or system failures, reinforcing the robustness of the proposed design.

Scenario ID	Test Case Description	Input Data	Expected Outcome
TS-01	**Ideal Match**	Req: A+, Dist < 5km	Donor selected & notified.
TS-02	**Distance Boundary**	Req: A+, Dist = 15km	Donor ignored (Out of range).
TS-03	**Blood Mismatch**	Req: A+, Donor: B+	Donor ignored (Incompatible).
TS-04	**Availability Check**	Donor Status: Busy	Donor excluded from list.
TS-05	**Empty Region**	No donors in 10km	System returns "No Donors Found" gracefully.

Table 4.1: Test Scenarios for Verification

Analysis of Verification Results

The verification results summarized in Table 4.1 demonstrate that the system correctly enforces all defined constraints across both standard and boundary scenarios. In test case TS-01, the system successfully identified and selected an eligible donor within the acceptable distance range, validating the correctness of spatial filtering and prioritization logic.

Test case TS-02 verifies the strict enforcement of distance thresholds. Donors located exactly at or beyond the defined boundary are excluded from selection, confirming precise reachability validation using exact distance computation rather than approximate filtering. This behaviour prevents false-positive donor selection during emergency coordination.

Blood group compatibility and donor availability constraints are validated through test cases TS-03 and TS-04, where incompatible or unavailable donors are consistently excluded from the candidate set. These results confirm that rule-based constraints are enforced prior to distance-based prioritization, maintaining selection correctness.

The system also demonstrates robust behaviour under absence-of-data conditions, as shown in test case TS-05. When no eligible donors are available within the specified region, the system returns a graceful response without failure or ambiguity. This behaviour is critical for real-world emergency systems where donor availability cannot be guaranteed.

Overall, the verification results confirm that the proposed system behaves deterministically and consistently across repeated executions, reinforcing its suitability for emergency blood donation coordination.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Performance Analysis

The performance of the proposed donor tracking and selection engine is evaluated based on its **accuracy, efficiency, determinism, and privacy preservation** in emergency blood donation scenarios. As the system is developed for **early-stage academic validation**, the evaluation emphasizes **functional correctness and algorithmic effectiveness** rather than large-scale stress testing or real-time deployment metrics.

The results indicate that the incorporation of **geospatial indexing techniques**, such as Geohash and Uber H3, substantially reduces the donor search space by restricting candidate evaluation to geographically relevant regions. By limiting distance computations to locality-based donor subsets, the system avoids exhaustive global scanning of the donor database, thereby improving computational efficiency and response readiness during emergencies.

The use of **exact distance computation through the Haversine formula** ensures accurate reachability validation. This prevents false-positive donor selections that may arise from approximate or purely radius-based filtering techniques. As a result, only donors who are geographically feasible are considered for final selection, improving the reliability of the coordination process.

Distance-based prioritization consistently resulted in the selection of the **nearest eligible donor** across all evaluated scenarios. This prioritization strategy is critical in emergency contexts, as reduced travel distance directly correlates with faster donor arrival and improved response outcomes. Furthermore, repeated executions of identical input scenarios produced the same ranked donor lists, confirming the **deterministic nature** of the proposed algorithm.

The evaluation also demonstrates that **privacy-first architectural decisions**, such as avoiding continuous real-time location tracking and employing spatial abstraction, do not negatively impact system performance. Spatial indexing provides sufficient geographic resolution for accurate donor matching while minimizing exposure of precise location data. This balance between performance and privacy enhances donor trust without compromising coordination effectiveness.

Overall, the performance analysis confirms that the proposed system achieves **accurate donor identification, efficient candidate reduction, deterministic behaviour, and privacy-aware operation**, making it suitable for emergency blood donation coordination in an academic and early implementation setting.

Feature	Traditional Manual System	Lifestream (Proposed System)
Search Method	Phone calls / Social Media Broadcasts	Automated Geospatial Querying
Response Time	High (Minutes to Hours)	Low (Milliseconds to Seconds)
Accuracy	Low (Messages sent to wrong city)	High (Only nearby donors notified)
Privacy	Low (Phone numbers shared publicly)	High (Location abstracted, data secure)
Scalability	Poor (Manual effort increases with users)	High (Database indexing handles load)

Table 5.1: Outcome Summary

5.2 Discussion of Results

The observed results confirm that modelling blood donation coordination as a **geospatial decision process** is more effective than traditional alert-based systems. The layered approach—combining spatial filtering, reachability validation, and prioritization—ensures both efficiency and correctness.

One of the key strengths of the proposed system is its **explainability**. Each donor selection decision can be traced back to clear constraints and distance calculations, which is critical for trust and validation in healthcare-related systems. The deterministic nature of the algorithm further simplifies verification and reduces ambiguity in emergency decision-making.

However, the results also highlight certain limitations. The absence of time-based reachability and real-world traffic conditions may affect donor arrival predictions in practical deployments. Additionally, the evaluation relies on simulated data rather than live hospital environments, which limits real-world generalizability.

Despite these limitations, the results demonstrate that the proposed system successfully addresses key gaps in existing blood donation platforms. The implementation provides a robust foundation for future enhancements while maintaining clarity, correctness, and privacy-awareness.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

This project presented *Lifestream*, a geospatially indexed mobile system designed to improve emergency blood donation coordination, with a focused implementation of its **Donor Tracking and Selection Engine**. The work addressed key limitations of existing blood donation platforms by moving beyond broadcast-based notifications and introducing a **deterministic, location-aware donor selection mechanism**.

The proposed system modelled donor coordination as a **multi-stage geospatial decision process**, combining spatial indexing, reachability validation, and distance-based prioritization. The use of **Geohash and Uber H3 indexing** enabled efficient reduction of the donor search space, while **Haversine-based distance computation** ensured accurate evaluation of donor reachability. Rule-based constraints such as blood group compatibility and donor availability further strengthened selection correctness.

The implementation and validation results demonstrated that the system consistently identifies eligible and reachable donors while preserving **deterministic behaviour and explainability**. Additionally, the adoption of a **privacy-first architectural approach** minimized unnecessary exposure of sensitive location data without compromising system functionality. Overall, the project successfully met its stated objectives and established a solid foundation for emergency blood donation coordination in an academic context.

6.2 FUTURE SCOPE

While the current work focuses on early implementation and validation, several enhancements can be explored in future iterations of the system:

- **Time-Based Reachability Analysis:**
  Incorporating travel-time estimation or isochrone-based reachability to improve donor arrival prediction.
- **Advanced Spatial Indexing:**
  Leveraging database-level H3 indexing or spatial query optimizations to support larger donor populations.
- **Intelligent Notification Strategies:**
  Introducing controlled and prioritized notification mechanisms to reduce donor fatigue.

- **Donor Wellness Integration:**
  Expanding the system to include non-clinical donor readiness checks and post-donation guidance.
- **Real-World Deployment and Evaluation:**
  Validating the system using live hospital data and real emergency scenarios.

These enhancements can be integrated without altering the core algorithmic structure, highlighting the **modularity and extensibility** of the proposed system.

REFERENCES

[1] H. Samet, *Foundations of Multidimensional and Metric Data Structures*, Morgan Kaufmann, 2006.

[2] Uber Technologies Inc., "**H3: A Hexagonal Hierarchical Spatial Index**," White Paper, 2018.
Available: https://h3geo.org/

[3] G. M. Morton, "**A Computer Oriented Geodetic Data Base and a New Technique in File Sequencing**," IBM Ltd., Ottawa, Canada, 1966.
(Foundational reference for Geohash concepts)

[4] C. Robusto, "**The Cosine-Haversine Formula**," *The American Mathematical Monthly*, vol. 64, no. 1, pp. 38–40, 1957.

[5] World Health Organization (WHO), "**Blood Donor Selection: Guidelines on Assessing Donor Suitability for Blood Donation**," WHO Press, 2012.

[6] S. Lee et al., "**Location-Based Service Systems: Issues and Challenges**," *IEEE Computer*, vol. 45, no. 3, pp. 80–85, 2012.

[7] A. Guttman, "**R-Trees: A Dynamic Index Structure for Spatial Searching**," *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pp. 47–57, 1984.

[8] Google Developers, "**Firebase Realtime Database Documentation**," Google LLC, 2023.
Available: https://firebase.google.com/docs/database

[9] M. Schiller and A. Voisard, "**Location-Based Services**," Morgan Kaufmann, 2004.

[10] J. Hightower and G. Borriello, "**Location Systems for Ubiquitous Computing**," *IEEE Computer*, vol. 34, no. 8, pp. 57–66, 2001.